

Geometry
Unit 2
Lesson 2 Practice

Name **Answer Key**
Date _____
Period _____

For questions 1-5, determine the type of reasoning for each statement.

1. For a golf cart to be considered 'street legal' it must have blinkers. Jill's golf cart has been deemed 'street legal'. Therefore, her golf cart has blinkers.

Deductive reasoning

2. When an object is thrown up in the air, it comes back down. Mark throws a baseball up in the air. The baseball will come back down.

Inductive reasoning

3. When Ethan stays up late, he is tired the next day. Ethan stayed up last night until 11:30 pm, therefore, Ethan will be tired the next day.

Deductive reasoning

4. The Olympic games are held every four years. The last games were held in 2020. The next games will be held in 2024.

Deductive reasoning

5. If a student passes an entrance examination, they will be admitted into their local community college. Barry passed the entrance examination. Barry will be admitted into the community college.

Deductive reasoning

For each statement in 6-10, write a counterexample that disproves the statement.

6. If a student attends the junior/senior prom, the student is either a junior or a senior. **Answers may vary. A sophomore attended the prom with a junior.**

7. If someone owns a cell phone, they are able to access the internet from it.

Answers may vary. To protect themselves, some law enforcement officials use cell phones that do not connect to the internet.

8. When two numbers are added, the result will always be a larger number.

Answers may vary. $-3 + -7 = -10$

9. The product of two odd numbers will be even.

Answers may vary. $7 \cdot 11 = 77$

10. If a number is divisible by 2, it is also divisible by 4.

Answers may vary. $\frac{18}{2} = 9$, $\frac{18}{4} = 4.5$, 18 is divisible by 2 and not by 4.